communications biology

A Nature Portfolio journal

Article

https://doi.org/10.1038/s42003-026-10671-5

Candidatus Bartonella xiongi influences iron homeostasis in the hematophagous mite *Dermanyssus gallinae*, affecting its reproduction

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Iron is an essential element for most organisms, including hematophagous arthropods, which are critical vectors for dozens of pathogens, seriously affecting the health of human beings and livestock. The mechanisms of iron acquisition by hematophagous arthropods remain incompletely characterized. Symbionts have established nutritionally reciprocal associations with arthropod hosts during long-term coevolution, endowing them with the potential to serve as an iron source. This study employed *Dermanyssus gallinae* (a notorious hematophagous mite) to explore the role of its key symbiont, *Bartonella*, in iron supply. A novel species, *Candidatus* Bartonella xiongi (strain DGB2), was isolated from the hematophagous mite. Genomic metabolic reconstruction showed that DGB2 encodes an *hmu* system responsible for heme uptake and degradation. Functional validation confirmed that DGB2 utilizes heme through its outer membrane heme receptor, HmuR, and electron donor-dependent heme oxygenase, HmuS. To investigate whether heme-derived iron was supplied to the host by DGB2, an antibiotic-induced dysbiosis model of *D. gallinae* with DGB2 depletion was established. DGB2 depletion significantly decreased the iron content within both mites and their eggs, as well as the ferritin expression levels, and compromised the reproduction of *D. gallinae*. Immunization of chickens with inactivated DGB2 vaccine elicited high-titer antibodies and significantly inhibited the fecundity of *D. gallinae*. Taken together, the present study demonstrates that the novel symbiont *Candidatus* Bartonella xiongi plays key roles in iron homeostasis of *D. gallinae* and contributes to the regulation of mite's reproductive capacity. Targeting symbionts related to iron homeostasis represents a promising biocontrol strategy for hematophagous vectors.

Iron, an essential element for nearly all living organisms[1], acts as a cofactor for diverse enzymes involved in core biochemical processes (e.g., DNA biosynthesis, oxygen transport, and energy metabolism)[2]. Among organisms, hematophagous arthropods (such as mosquitoes, ticks, bedbugs, and some mites) play crucial roles in the transmission of dozens of deadly pathogens (such as *Plasmodium*, dengue fever virus, and *Babesia*) to human beings and livestock[3–5]. For obligate hematophagous arthropods, iron is critical for sustaining normal physiology, particularly reproduction and embryonic development[6]. For example, in the hematophagous pest *Aedes aegypti* (mosquito), iron is transported to the ovaries and eggs to participate in reproductive processes[7,8]. An iron-deficient diet or RNA interference targeting ferritin severely compromises fecundity and egg development in ticks, mosquitoes, and mites[9–11].

Given their obligate hematophagy, most hematophagous arthropods rely exclusively on host blood for iron, which exists primarily as ferric-transferrin and hemoglobin[12], but their iron acquisition route from host

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blood remains ambiguous. Iron from transferrin exhibits typically higher bioavailability[8] and was found to be related to supporting embryonic development in *A. aegypti* and *Ixodes scapularis* (tick)[13]; and it was even considered as a potentially major iron source in ticks[13]; however, this hypothesis still lacks direct evidence and faces at least two key issues: i) In mammary serum, the amount of iron bound to transferrin under normal physiological conditions is very low. For example, the mean value was 1.2 ng Fe/mL in human blood[14], and transferrin binds ferric ions (two Fe^{3+}) with an extraordinarily high affinity ($Kd = 10^{-23}M$)[15]. It remains unclear how ticks can efficiently extract enough iron from host transferrin with such scarce and tightly bound iron. ii) In ticks, ferritins have been confirmed to act as key iron transporters and storage proteins. These multimeric complexes consist of heavy (Ferritin Heavy Chain, FTH) and light (Ferritin Light Chain, FTL) chains, with FTH possessing ferroxidase activity that oxidizes reactive ferrous (Fe^{2+}) to ferric (Fe^{3+}) iron, subsequently forming inert ferrihydrite aggregates within ferritin[11]. It remains unclear how tick ferritins mediate the transport of ferric iron—rather than ferrous iron—originating from host transferrin. Therefore, further research is needed to clarify whether hematophagous arthropods acquire iron directly from transferrin. Besides transferrin, about 80% of the iron in vertebrate blood occurs as heme in hemoglobin[16], making heme a more ideal iron source for hematophagous arthropods. In fact, most animals studied to date (including several hematophagous arthropods) acquire iron via heme degradation. For example, iron source tracing and quantification in *A. aegypti* have firmly established heme as the primary iron source supporting reproduction (with heme-derived iron accounting for 98% of the total iron in eggs and 97% in adult females)[8]. Although iron is predominantly present in heme, heme-bound iron requires enzymatic degradation for bioavailability in hematophagous arthropods[6], a reaction catalyzed by heme oxygenase (HO) to yield ferrous ions (Fe^{2+}), carbon monoxide (CO), and biliverdin (BV) IXα[17]. Unexpectedly, the function of HOs in most hematophagous arthropods has not been definitively confirmed. Two scenarios prevail in the HOs of hematophagous arthropods: i) the complete absence of HO-encoding genes in the genomes of Acari representatives, such as *I. scapularis* and *D. gallinae*[13,18]; ii) the presence of HOs, whose functions either remain unconfirmed (as in *Rhodnius prolixus*)[19,20] or carry out non-canonical roles[21,22]. Notably, although HO is present in *Anopheles gambiae*, it shows extremely low catalytic activity, making it ineffective at mediating heme degradation[21]. Moreover, in *A. aegypti*, the spatiotemporal expression profile of HO points to a primary role in antioxidant stress responses rather than supporting iron metabolism through heme breakdown[22]. Overall, the pathways by which hematophagous arthropods acquire iron via heme degradation remain to be elucidated, and alternative uncharacterized heme degradation strategies may exist.

Interestingly, to overcome host-imposed iron limitation, diverse bacterial taxa have developed specialized and efficient heme uptake and degradation systems for heme-derived iron acquisition[23]. Among them, Gram-negative bacteria (such as *Mycobacterium tuberculosis* and *Campylobacter jejuni*) typically employ outer membrane receptors to capture heme and deliver it to the periplasm in a TonB-dependent manner[24]. Subsequently, it is translocated into the cytoplasm via inner-membrane ABC transporters, where it is catalytically degraded by HO, releasing iron ions to support bacterial growth requirements[25]. Meanwhile, to maintain intracellular iron homeostasis, supraphysiological amounts of iron are efficiently effluxed by bacteria through multiple systems, such as P-type ATPases, cation diffusion facilitators (CDFs), and major facilitator superfamily (MFS)[26]. Hematophagous arthropods harbor complex and highly abundant symbiotic bacteria[27]. Thus, theoretically, symbiotic bacteria have the potential to supply iron to hematophagous arthropods by uptaking and degrading heme and releasing iron, yet this putative mechanism remains unvalidated and unexplored.

It is noteworthy that complex symbiotic bacterial communities reside stably in hematophagous arthropods, and an intimate mutualistic relationship focused on nutrient supply has been established through prolonged coevolution. Symbionts, such as *Wolbachia* (*w*Cle)[28], *Francisella*[29], and *Wigglesworthia*[30] are capable of synthesizing B vitamins to remedy the nutritional deficiencies of their hosts, thereby supporting critical physiological processes, including development and reproduction. Moreover, symbionts contribute to amino acid synthesis and metabolism, thus regulating host feeding behavior and reproductive fitness[31–33]. On this basis, it is reasonable to speculate that the efficient heme uptake and degradation systems of symbionts may serve as an alternative pathway for hosts to obtain the essential metal element, iron, especially when the endogenous heme degradation pathways have not been elucidated. In fact, *Sodalis glossinidius* in tsetse flies[34] and *Bartonella henselae* in fleas[35] possess the ability to obtain heme-derived iron from their hosts, but whether this iron can be fed back to the hosts remains unexplored.

D. gallinae is the most devastating hematophagous ectoparasite in poultry. In addition, *D. gallinae* has been reported to carry a broad spectrum of zoonotic pathogens, including *Salmonella* spp., *Borrelia burgdorferi*, and avian influenza virus, which can be mechanically or biologically transmitted to susceptible hosts through its blood-feeding, presenting considerable public health threats[36]. In the symbiotic microbiome of *D. gallinae*, *Bartonella* is the predominant genus, which can be divided into two groups: *Bartonella* Group A and *Bartonella* Group B[37,38]. The former was identified in our previous study[39] and named *Bartonella kongi* strain DGB1 (GenBank accession number: GCA_041345015.1). In addition, *Kocuria massiliensis* was also detected as another dominant species[39]. However, compared with *K. massiliensis*, *Bartonella* is deemed to contribute significantly to the mite's biological processes due to its high abundance and prevalence[37,38], yet its refractoriness in vitro isolation has greatly limited further exploration of its functional roles. The present study successfully isolated the key symbiont *Bartonella* sp. DGB2 (previously classified as *Bartonella* Group B) from *D. gallinae*, finding that it functions in mite iron homeostasis, potentially serving as one of the iron sources, and modulates mite fitness. Intervention against DGB2 in mites via antibiotic treatment disrupts iron supply, and application of an inactivated vaccine targeting this key symbiont related to iron homeostasis provides an effective approach to control *D. gallinae*. Herein, this study revealed a novel pathway involving symbiotic bacteria in maintaining iron homeostasis in hematophagous arthropods and further demonstrated the feasibility of biological control against hematophagous pests by targeting symbionts involved in iron homeostasis.

Results

Isolation and characterization of a novel species of the genus *Bartonella* (strain DGB2) from *D. gallinae*

A strain of *Bartonella* was isolated from homogenates of *D. gallinae*, and its 16S *rRNA* gene sequence exhibited 99.78% similarity to *Bartonella* Group B (accession: MF086616.1) reported by Hubert J et al.[37], which was designated as strain DGB2 in the present study. Colonies grown on TSA blood agar plates were round, smooth, and translucent (Fig. 1A). TEM inspection revealed that DGB2 cells presented a rod-shaped morphology, with an average dimension of approximately 1.5 μm in length and 0.6 μm in width ($n = 50$), and featured a double-membrane structure along with a single polar flagellum (Fig. 1B, Supplementary Fig. 1).

The genome of strain DGB2 was de novo assembled using PacBio sequencing, revealing an estimated size of 2,249,190 bp and comprising a chromosome alongside four plasmids (Fig. 1C, Supplementary Table 1). Phylogenetic relationships based on whole-genome sequences of strain DGB2 with other *Bartonella* isolates (listed in Supplementary Data 1) revealed that DGB2 clustered within a distinct species cluster (Supplementary Fig. 2). Whole-genome DDH and ANI values of strain DGB2 were calculated against phylogenetically related *Bartonella* species (Supplementary Table 2), with DDH values ranging from 19.3% to 41.9% and ANI values from 67.51 to 68.29% (Supplementary Table 3, Supplementary Figs. 3 and 4). With 70% DDH and 95% ANI established as the universal thresholds for species delineation[40,41], the fact that both values of strain DGB2 fall below these cut-offs strongly supports its classification as a novel *Bartonella* species. This strain has been deposited in the China General Microbiological Culture Collection Center (collection number: CGMCC

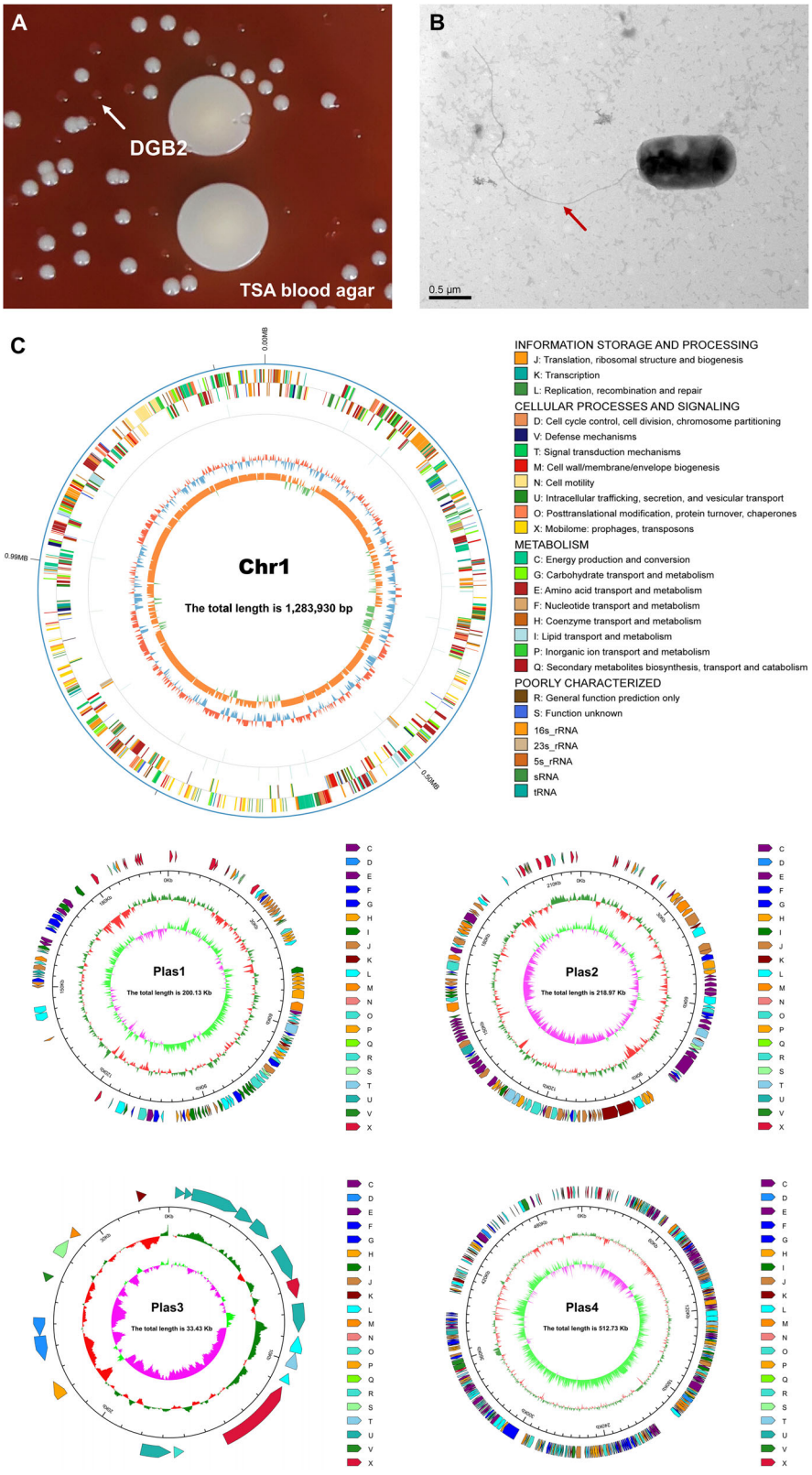


Fig. 1 | Morphological observation and circular genome representation map of strain DGB2. **A** DGB2 colonies (white arrow) grown on TSA blood agar. **B** Transmission electron micrographs of negatively stained cells of DGB2 isolates. Flagella-like structures were indicated with red arrows. Scale bars indicated 0.5 μm. **C** Circular genome representation map of DGB2. From outside to the center in Chr1: genome sequence of coordinates, genes on forward strand (color by COG categories), genes on reverse strand (color by COG categories), ncRNA, GC content, GC skew. From outside to the center in Plas1-4: COG functional annotation of classified genes (arrows in clockwise direction indicate coding on the positive strand), genome sequence of coordinates, GC content, GC skew. bp base pairs.

No.36715). We propose designating this newly identified species as *Candidatus* Bartonella xiongi, in honor of the eminent Chinese parasitologist Prof. Dashi Xiong.

Potential metabolic functions of DGB2 for nutrient provision

GeneMarkS analysis of the initial assembly of strain DGB2 identified a total of 2537 protein-coding genes with an average size of 787 bp, covering 88.81% of the genome. Subsequently, the nutrient-related metabolic pathways of DGB2 were reconstructed based on the functional annotation of protein-coding genes. The DGB2 genome encodes conserved genes for diverse B vitamin and cofactor biosynthesis, yet the integrity of individual biosynthetic pathways varies considerably. The synthesis pathways of riboflavin (vitamin B2), pantothenate (vitamin B5), and biotin (vitamin B7), along with the cofactors FAD and CoA, are entirely preserved (Fig. 2A, B).

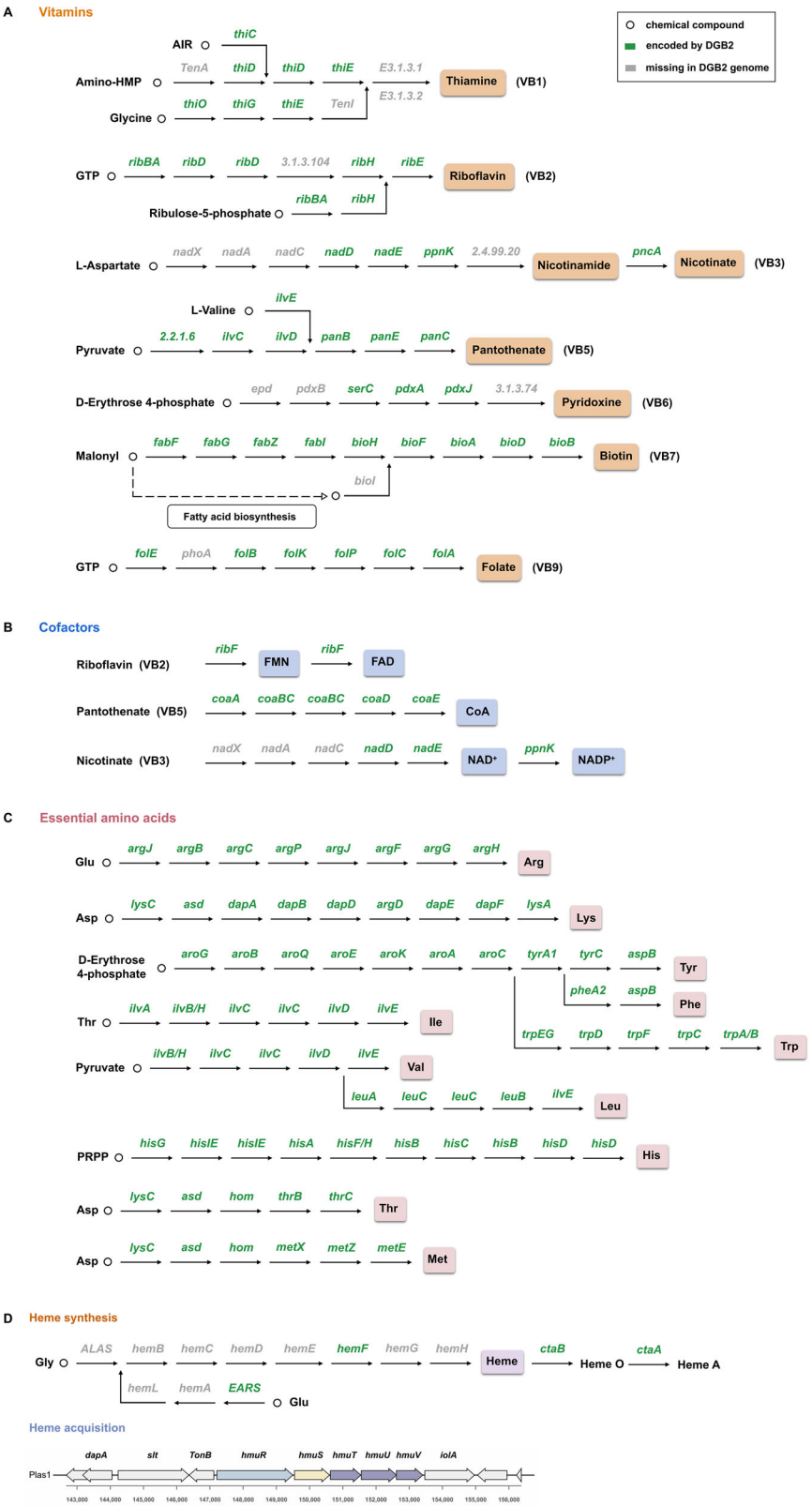


Fig. 2 | Metabolic reconstruction of nutrient biosynthesis and heme utilization pathways. **A** Vitamins, **B** Cofactors, **C** Essential amino acids, and **D** Uptake and degradation of heme. Circles and arrows represent compounds and genes, respectively, in synthetic pathways. Green shows genes presented in DGB2 genome. Missing genes were shown in gray.

Additionally, the biosynthetic pathway of folate (vitamin B9) remains substantially intact (6/7 genes present), with the absence of gene *phoA*, which might be compensated by *folE* gene[42] or be entirely dispensable for folate biosynthesis[43]. In contrast, the loss of numerous genes severely restricts strain DGB2's capacity to supply the host with thiamine (vitamin B1), nicotinate (vitamin B3), pyridoxine (vitamin B6), and NAD(P) (Fig. 2A, B). Thus, given the importance of B vitamins and cofactors in hematophagous arthropods and limited source from host blood, DGB2 likely contributes to mite fitness by supplying these scarce blood-borne nutrients.

Metabolic reconstruction of amino acids biosynthesis pathways revealed that DGB2 encodes all genes required for the biosynthesis of essential amino acids (EAAs) (Fig. 2C), including Lys, Phe, Trp, Ile, Val, Leu, His, Thr, Met, as well as two conditionally EAA (Arg and Tyr). Compared

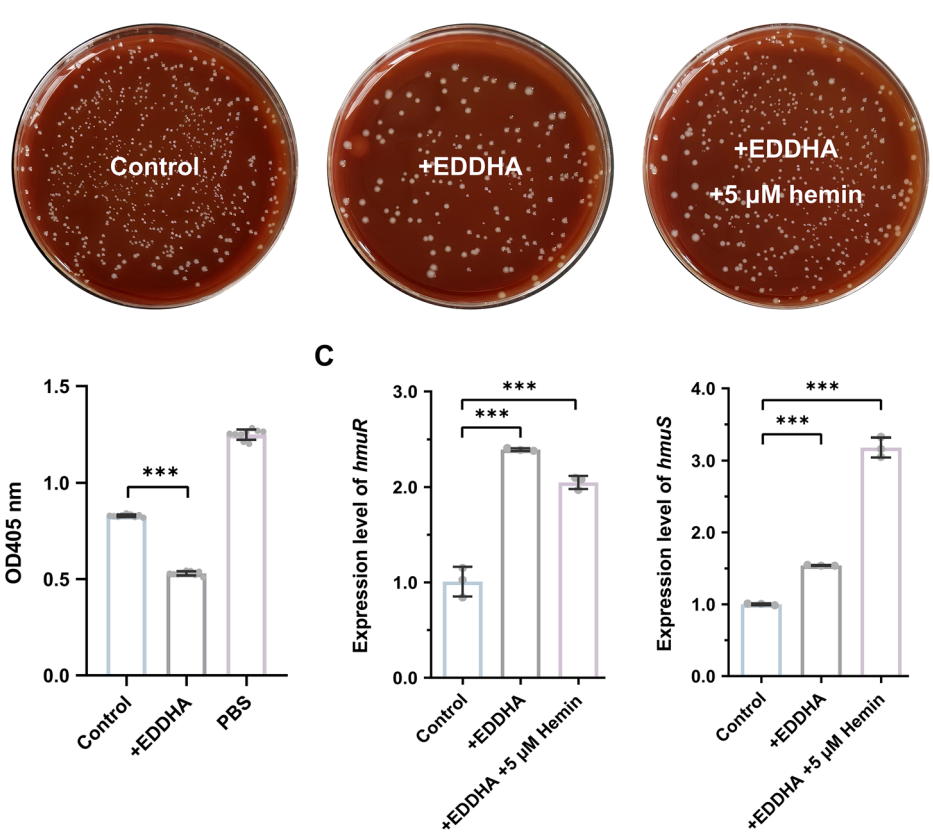


Fig. 3 | Physiological and molecular responses of DGB2 under iron deficiency stress. A Growth status of DGB2 under iron-depleted conditions or supplemented with 5 μM hemin as an iron source. Untreated S10 medium served as the control. **B** The heme-binding capacity of DGB2 under iron deficiency stress. Control: Untreated DGB2 cells; +EDDHA: DGB2 cells treated with 100 μM EDDHA; PBS: buffer supplemented with working-concentration hemin (no DGB2 cells included, serving as blank control) ($n = 9$). **C** The transcriptional levels of *hmuR* and *hmuS* of DGB2 under iron-depleted conditions or supplemented with 5 μM hemin as an iron source ($n = 3$). Data are presented as mean ± SD. Error bars represent SD. Significance was determined by one-way ANOVA. ***$p < 0.001$.

with endosymbionts in other hematophagous arthropods[44], such substantial variability may endow strain DGB2 with a unique capacity to provide EAAs to hematophagous arthropod hosts.

Analysis of the heme metabolic pathway revealed that the DGB2 genome is deficient in numerous crucial genes within the heme biosynthetic pathway (Fig. 2D), suggesting that the bacterium does not synthesize heme de novo but utilizes heme from the mite intestine, which was corroborated by further analysis of the heme uptake and degradation system in DGB2 genome. As shown in Fig. 2D, an *hmuRSTUV* cluster was found on Plasmid 1 in DGB2, which encodes a TonB-dependent ABC transporter system for the uptake and utilization of heme, whose function has been demonstrated in *Yersinia* spp.[45]. The outer membrane heme receptor encoded by *hmuR* captures heme, which is transported to the cytoplasm via the *hmuTUV* complex and degraded by the *hmuS*-encoded HO to release iron ions. Comparative analysis showed that the heme receptor HmuR (WP_412058372.1) and HO HmuS (WP_412058371.1) of strain DGB2 share 23.2% and 38% amino acid similarity with the corresponding proteins in *Yersinia enterocolitica* (ATCC 9610), respectively, indicating that DGB2 possesses the potential to uptake and utilize heme as an iron source. In contrast, the two other dominant bacteria in mites, *Bartonella* sp. DGB1 and *K. massiliensis*, exhibit substantial gene losses in their heme transport systems (Supplementary Fig. 5), which basically eliminates the possibility of these two bacteria utilizing heme as an iron source for the mite host.

Utilization of hemin as an iron source under iron scarcity

To verify hemin acquisition by DGB2, an iron-deficient environment was established by supplementing S10 medium with the iron chelator EDDHA, resulting in significant growth inhibition of strain DGB2 (Fig. 3A). Nevertheless, this growth impairment was rescued by exogenously supplemented hemin (Fig. 3A), validating that under iron-deficient conditions DGB2 is capable of acquiring hemin as an iron source for its proliferation. This finding was further corroborated by the markedly enhanced hemin-binding capacity in strain DGB2 subjected to iron starvation stress (Fig. 3B).

Transcriptional levels of *hmuR* and *hmuS* were determined using *rplD* as a reporter gene under iron-limiting conditions. Results demonstrated that, following EDDHA treatment, the expression levels of *hmuR* and *hmuS* were notably upregulated by 2.39 and 1.54 folds, respectively (Fig. 3C). Upon exogenous supplementation of 5 μM hemin as an iron source, the up-regulation of *hmuS* expression was further enhanced (Fig. 3C), verifying the function of the *hmu* system in hemin acquisition by DGB2 under iron-limiting conditions.

Heterologous expression of HmuR in *E. coli* confer the capacity to bind an increased amount of hemin

Considering that HmuR exhibits low sequence similarity to other well-characterized heme receptors, heterologous expression in *E.coli* BL21 (DE3) was performed to verify the hemin-binding capacity of HmuR. Overexpression of the:: HmuR::His_6 fusion protein in this strain was confirmed via sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE), and its molecular weight was consistent with the expected size (89.2 kDa) (Fig. 4A). It was found that overexpression of:: HmuR::His_6 conferred enhanced hemin-binding capacity on *E.coli* (Fig. 4B), demonstrating that HmuR is a functional protein mediating heme acquisition in strain DGB2.

HmuS functions as heme oxygenase, catalyzing hemin degradation in vitro

To validate the function of HmuS, recombinant HmuS was obtained through overexpression followed by Ni-NTA affinity chromatography, and a single band at approximately 38 kDa was detected by SDS-PAGE, which was consistent with the predicted molecular weight (Supplementary Fig. 6). Subsequently, the binding of HmuS to hemin was assayed via absorption spectroscopy, showing that free hemin displayed a Soret band at ~385 nm, whereas in the presence of rHmuS, this absorption peak experienced a shift to ~410 nm, leading to the distinctive Soret peak of the hemin-protein complex (Fig. 4C). In addition, it was observed that the Soret peak at ~410 nm gradually increased in intensity as increasing amounts of hemin

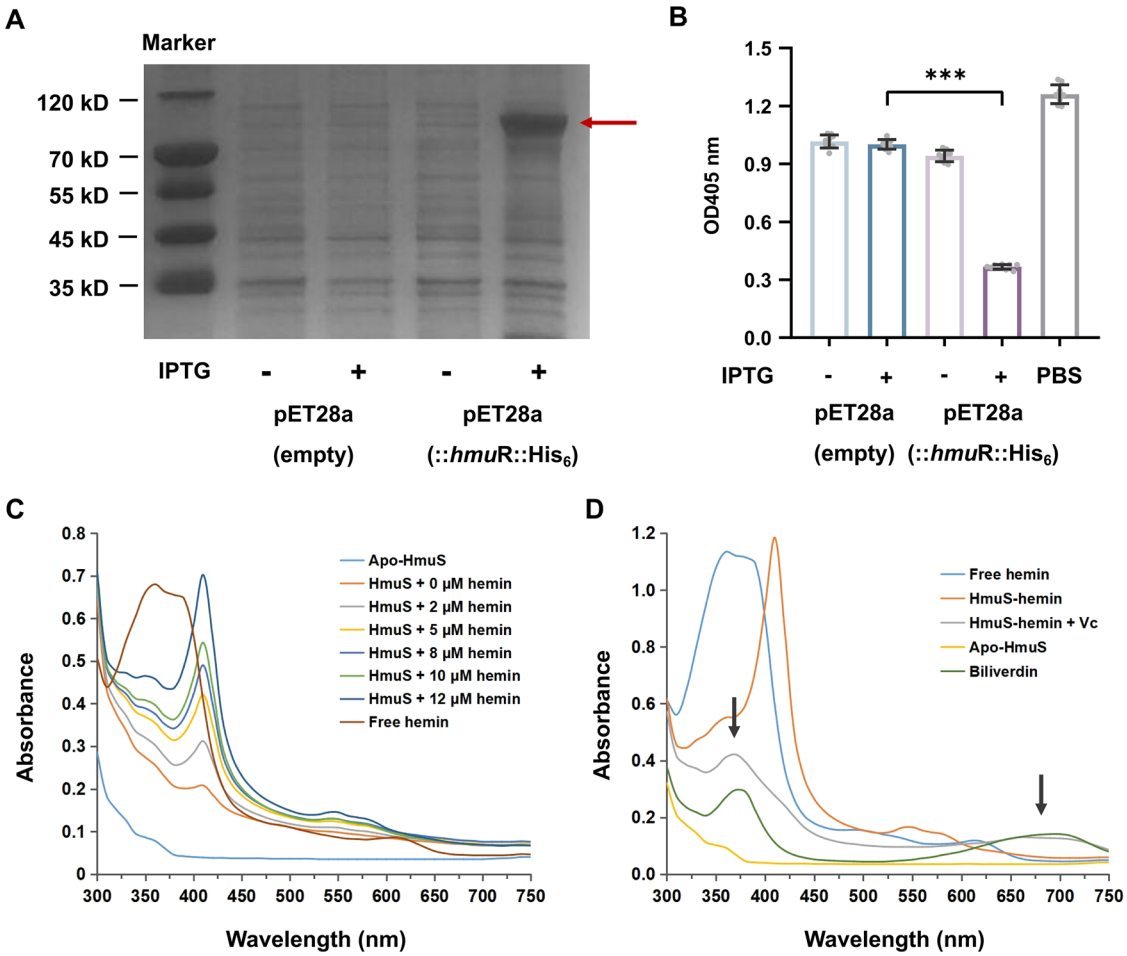


Fig. 4 | Functional analysis of HmuR and HmuS in hemin uptake and degradation. A SDS-PAGE analysis of recombinant HmuR expression: pET28a (empty vector) or pET28a::*hmuR*::His_6 were induced (+) or uninduced (-) with IPTG. **B** Hemin-binding capacity of *E. coli* heterologously expressing *hmuR*. *E. coli* BL21 (DE3) cells contained pET28a (empty vector) or pET28a::*hmuR*::His_6 were induced (+) or uninduced (-) with IPTG. PBS: buffer supplemented with working-concentration hemin (no bacterial cells included, serving as blank control) ($n = 9$). Data are presented as mean ± SD. Error bars represent SD. Significance was determined by one-way ANOVA. ***$p < 0.001$. **C** Binding of hemin to HmuS. Increasing amounts of hemin (0-12 μM final concentration) were added to HmuS (20 μM), and the spectrum (300–750 nm) was recorded. **D** HmuS dependent degradation of hemin. 5 mM final concentration of ascorbic acid was added to the HmuS-hemin complex. The spectral changes at 300–750 nm were monitored after 30 min of reaction; arrows indicate the absorption peaks of the hemin degradation product. Experiments were performed in triplicate, and a single representative experiment is presented.

were added to rHmuS, demonstrating a concentration-dependent manner (Fig. 4C).

In the presence of ascorbic acid (as the electron donor), hemin was degraded by rHmuS, characterized by the disappearance and shift of the Soret peak at ~410 nm, along with the emergence of a new product spectrum with broad maxima at 380 nm and 680 nm which was very similar to those of biliverdin standard (Fig. 4D). Combined with its absorption properties[46] and the high consistency of characteristic peaks with the biliverdin standard, the product was presumed to be iron-free biliverdin, but further confirmation is needed by other techniques, such as MS. Collectively, these results support the conclusion that DGB2 can acquire iron via HmuS degrading hemin.

DGB2 encodes an MbfA-type ferrous ion efflux pump

To assess whether heme iron acquired by DGB2 can be supplied to the host, we identified a ferrous ion efflux pump (WP_008037385.1) based on the functional annotation results of protein-coding genes. The multiple sequence alignment and phylogenetic analysis revealed that it is closely related to the membrane-bound ferritin (MbfA)-type ferrous efflux transporter from *Rhizobium* spp. and shares the same category of conserved domains (Supplementary Fig. 7A). Consistent with the known MbfA-type efflux pumps in *Agrobacterium Tumefaciens*, the predicted secondary structure of this transporter contains five putative transmembrane domains and the putative iron-binding motifs E-6-Y and ExxH (Supplementary Fig. 7B), indicating its potential to efflux ferrous ions. To further verify its function, we attempted to knockout the encoding gene via a suicide plasmid[47] with assistance from the group of Prof. Yuan, but failed. As an alternative, we subsequently implemented the selective elimination of DGB2 in mites to validate the strain's contribution to iron supply.

Inhibition of DGB2 reduced the reproductive capacity of *D. gallinae*

To verify whether iron acquired by DGB2 can be supplied to the host, DGB2 was specifically eliminated by treating *D. gallinae* adults with AMX. Five days post-treatment with AMX at 180 mg/kg, the abundance of DGB2 in adult females was significantly reduced compared to the control group, demonstrating a more pronounced inhibitory effect of AMX on this bacterium. The other dominant symbionts, including DGB1 (designated *Bartonella* A in our earlier study[39]) and *Kocuria*, also showed slight abundance fluctuations due to the broad-spectrum activity of the antibiotic, but with no statistically significant differences (Fig. 5A). Interestingly, AMX treatment exhibited a strong inhibitory effect on mite reproduction (Fig. 5B). In comparison to the control group, a higher number of adult females experienced oviposition defects after AMX treatment, i.e., failure to lay eggs

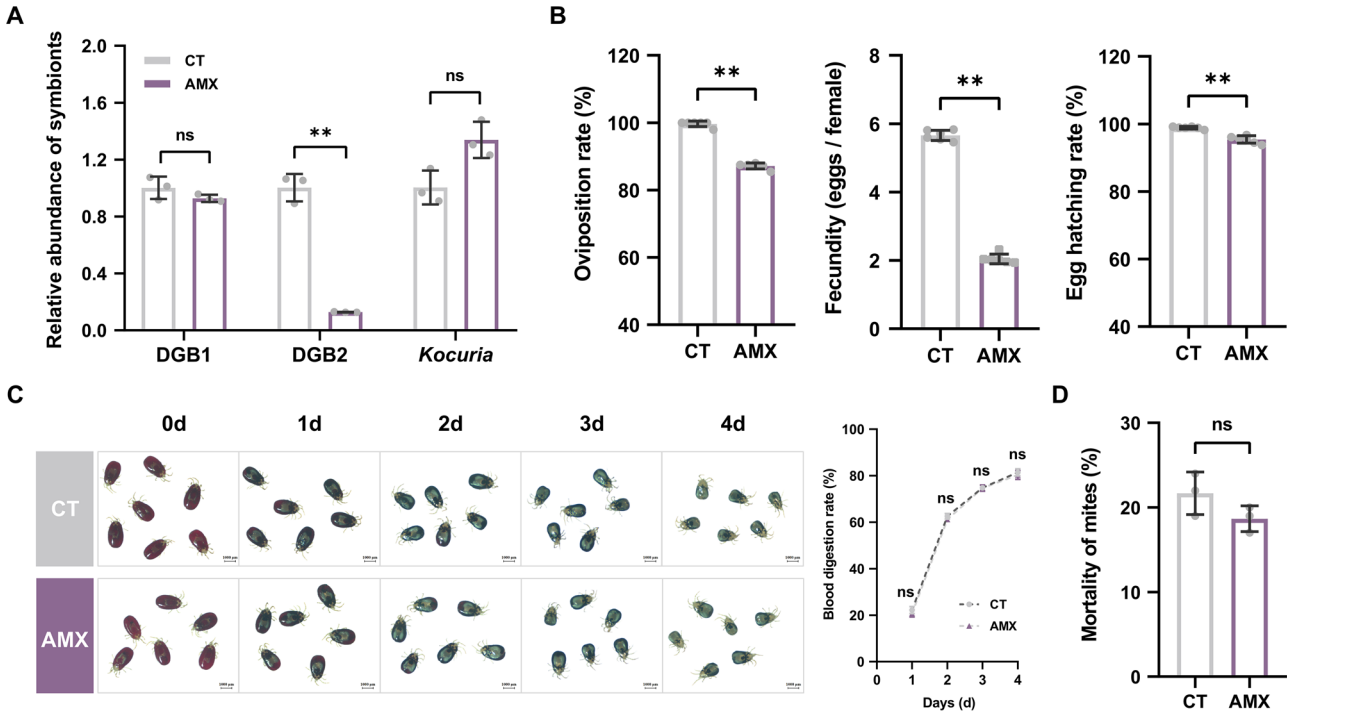


Fig. 5 | Effects of AMX treatment on symbiont abundance and physiology in *D. gallinae*. **A** Relative abundance of *Bartonella* sp. DGB1, DGB2, and *Kocuria* in adult females treated with 0.9% NaCl (CT) or amoxicillin (AMX) ($n = 3$). **B** Effects of AMX treatment on reproduction of *D. gallinae*. Oviposition rate, fecundity, and egg hatching rate were determined within 5 days ($n = 6$). **C** Effects of AMX treatment on blood digestion. Appearance observation and weighing were carried out daily for 4 consecutive days ($n = 3$). Scale bars indicated 1000 μm. **D** Effects of AMX treatment on survival. Mites were deemed dead in the absence of active movement of legs evoked by either light or mechanical stimulation. The cumulative mortality rate of mites over a 5-day period was calculated ($n = 3$). Data are presented as mean ± SD. Error bars represent SD. Significance was determined by Student's *t* test or Mann-Whitney test. $^{**}p < 0.01$, ns indicates no significant difference.

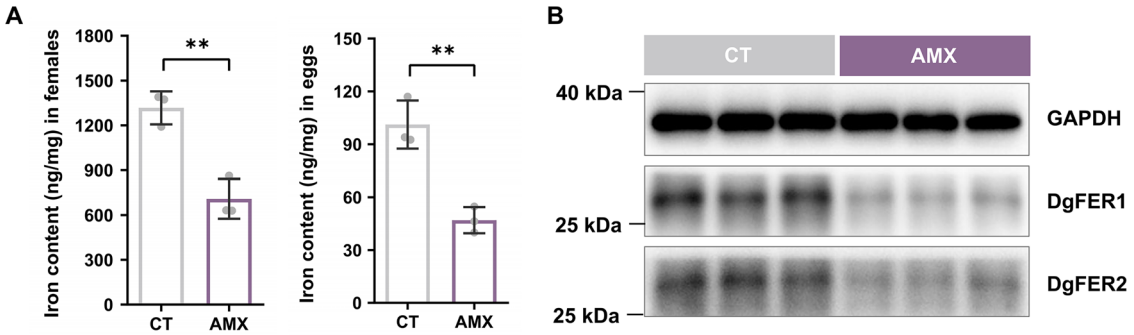


Fig. 6 | Impact of AMX on bioavailable iron levels in mites. **A** Iron content in AMX-treated adult females and their eggs. Mites on day 7 post-feeding and their eggs were collected to quantify iron content by ICP-MS. The ICP-MS readings were expressed as ng Fe per mg of mites or eggs ($n = 3$). Data are presented as mean ± SD. Error bars represent SD. Significance was determined by Student's *t* test. $^{**}p < 0.01$. **B** Ferritin expression level detection in adult females ($n = 3$). DgFER1: secreted ferritin; DgFER2: intracellular ferritin; GAPDH: protein reference.

normally, with the average oviposition rate decreasing from 99.66 to 87.15%. Furthermore, a significant reduction in mite fecundity was clearly observed in the AMX group (Fig. 5B), with the average number of eggs laid per adult decreasing from 5.66 to 2.05, indicating a deficiency of nutrients required for reproduction. Additionally, a significant decrease in egg hatching rate was also observed compared to the control group, dropping from 99.88 to 95.41% (Fig. 5B). However, both appearance observation and weighing assays confirmed that AMX-mediated DGB2 inhibition did not affect mite blood digestion (Fig. 5C). Consistently, no discernible impact of AMX exposure on the survival capacity of the mites was observed (Fig. 5D).

DGB2 contributes to iron supply for *D. gallinae*

Previous studies have demonstrated that blood within mites is almost entirely digested by day 6, or even earlier, following a blood meal[48,49]. Iron content was measured in mites at different time points of blood digestion and found that iron concentrations remained relatively stable from day 7 post-blood feeding (Supplementary Fig. 8), at which point the interference from host hemoglobin-derived iron could be considered negligible. Afterward, the mites on day 7 post-feeding and their eggs were collected to quantify the iron content. Results demonstrated that suppression of DGB2 via AMX treatment led to a significant reduction in iron levels in adult females and their progeny eggs (Fig. 6A). Specifically, the iron content in adult females decreased by 46.22% (from 1317.28 ng/mg to 708.41 ng/mg), while that in their laid eggs was reduced by 53.51% (from 101.18 ng/mg to 47.04 ng/mg). Interestingly, AMX treatment significantly reduced the expression levels of the secreted ferritin DgFER1 (involved in iron transport and referred to as FER2 in ticks) and the intracellular ferritin DgFER2 (responsible for iron storage and referred to as FER1 in ticks), reflecting

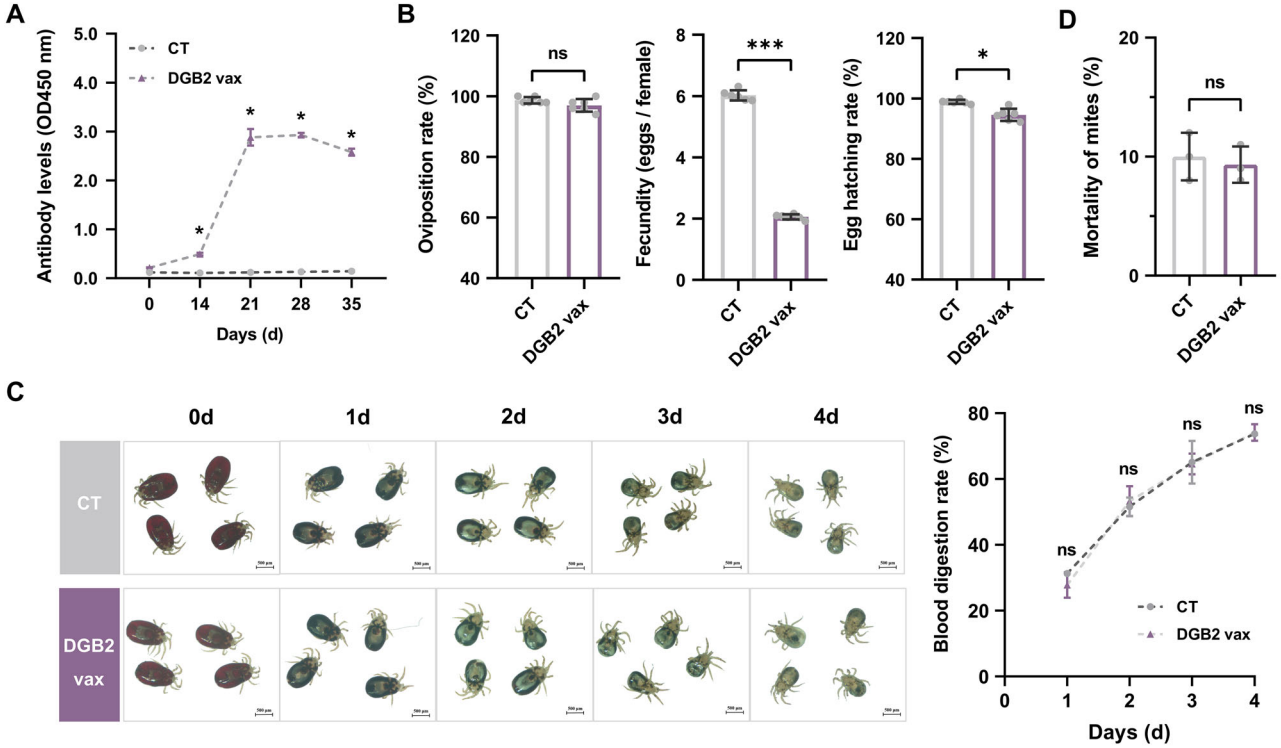


Fig. 7 | Efficacy of inactivated DGB2 vaccines. A Antibody response in immunized chickens. Serum antibody titers to DGB2 were determined by ELISA and expressed as the OD450 value ($n = 3$). Antibody titers were compared between immunization and control chickens by Student's t test. *$p < 0.05$. Impact of inactivated DGB2 vaccine on *D. gallinae* **B** reproduction ($n = 6$), **C** blood digestion ($n = 3$), scale bars indicated 500 μm, and **D** survival ($n = 3$). Data are presented as mean ± SD. Error bars represent SD. Student's t test or Mann-Whitney test was utilized for assessing differences. *$p < 0.05$, ***$p < 0.001$, ns indicates no significant difference.

decreased bioavailable iron levels and disturbed iron homeostasis within mites (Fig. 6B). These findings suggest that symbiont DGB2 contributes to the maintenance of iron homeostasis, especially in processes related to iron supply in *D. gallinae*.

Immunization with inactivated DGB2 vaccine affords effective protection against *D. gallinae*

Substantial levels of antigen-specific IgY were induced in chickens by immunization with the inactivated DGB2, peaking after the third immunization (Fig. 7A). Compared to the adjuvant group, the fecundity of *D. gallinae* was significantly reduced in the immunized group (Fig. 7B), with an average number of eggs laid per adult decreasing from 6.02 to 2.05, accompanied by a decline in the egg hatching rate (dropped from 98.90 to 94.57%). No significant effects were observed on blood digestion (Fig. 7C) or survival (Fig. 7D). These results are consistent with the performance changes observed in response to AMX treatment. These findings demonstrate that immunization with the inactivated DGB2 confers favorable protective efficacy for the management of *D. gallinae*.

Discussion

HO-mediated heme degradation represents an important source of iron in many organisms. However, the absence or functional limitation of HOs in hematophagous arthropods renders the pathways for heme iron acquisition still ambiguous. It has been demonstrated that some key symbionts in hematophagous arthropods possess the capacity to acquire heme iron from their arthropod hosts, such as *S. glossinidius* in tsetse flies[34] and *B. henselae* in fleas[35]. Given the highly specialized nutritional mutualism existing between symbiotic bacteria and their arthropod hosts, these findings collectively indicate the existence of a putative symbiont-mediated alternative pathway for heme iron acquisition, which may represent a critical adaptive strategy for hematophagous arthropods.

As a typical representative of hematophagous arthropods, *D. gallinae* harbors a complex bacterial community with *Bartonella* as the core genus[37,38]. Among the two distinct *Bartonella* species, the prevalence of *Bartonella* sp. A across *D. gallinae* populations was 100%, with that of *Bartonella* sp. B at 97.2%; both rates were substantially higher than those of other symbiotic bacteria, including *Cardinium* (52.8%), *Wolbachia* (36.1%), and *Rickettsiella* (1.4%)[38]. Moreover, these two *Bartonella* species displayed notable dominance in abundance, with the maximum individual relative abundance of *Bartonella* sp. A reaching up to 95.83% and that of *Bartonella* sp. B reaching 77.80%[38]. In hematophagous arthropods, dominant abundance has been proven to be essential for playing key roles in host physiology. For instance, *Francisella* in *Ornithodoros moubata* (African soft tick), *Wigglesworthia* in *Glossina morsitans morsitans* (tsetse fly), and *Wolbachia* in *Cimex lectularius* (bedbug) all play crucial roles in host development, survival, and reproduction and have high abundance[29,30,50]. Therefore, the high abundance and prevalence of *Bartonella* in *D. gallinae* suggest potentially critical roles in the mite's physiology, which have not been confirmed by trials. In this study, a novel *Bartonella* strain (designated DGB2) was isolated from *D. gallinae*, with its 16S *rRNA* gene sequence showing >97% similarity to both *Bartonella* group B (MF086616.1) identified in the Czechia mite population and *Bartonella* sp. B-YN-2019 (LC710640) found in the Japanese mite population. Comparative genomic analysis based on DDH and ANI values confirmed that this strain represents a novel species of the genus *Bartonella*, hereby designated as *Candidatus* Bartonella xiongi. Isolation of this strain not only enriches the taxonomic diversity of the genus *Bartonella*, but also provides critical material support for subsequent functional investigations.

During coevolution with their hematophagous arthropod hosts, symbiont genomes generally undergo streamlining, but retain genes for synthesizing blood-scarce nutrients (e.g., B vitamins and cofactors)[29,51]. Generally, these symbiont-derived nutrients are indispensable for sustaining key physiological processes, including development and reproduction in

hematophagous arthropods[28–30]. Metabolic pathway reconstruction in the DGB2 genome demonstrated the presence of complete biosynthetic pathways for multiple B vitamins (riboflavin, pantothenate, biotin) and cofactors (FAD, CoA). In hematophagous arthropods, symbionts with confirmed B-vitamin biosynthetic capacity also encode similar pathways to DGB2[44,52,53], suggesting that DGB2 may perform a comparable nutritional symbiotic function by supplying these essential nutrients to promote the reproductive fitness of the mite. Interestingly, the reconstruction of essential amino acid metabolic pathways revealed an unexpected result: DGB2 encodes all the genes for essential amino acid biosynthesis, which are generally degenerated in other hematophagous arthropod symbionts[39,44], as blood contains sufficient essential and non-EAAs to satisfy nitrogen requirements[54]. A reasonable explanation is that, as a facultative intracellular bacterium, *Bartonella* sp. DGB2 may be in the early stages of transition from a free-living to an intracellular lifestyle, thus lacking significant genome reduction and loss of metabolic function compared with obligate intracellular symbionts (e.g., *Rickettsia* spp. and *Coxiella* spp.)[55]. In fact, a growing body of evidence indicates that in arthropods like ticks and aphids, facultative symbionts have evolved into beneficial associates, even entirely replacing the ancestral primary symbionts and conferring hosts with novel metabolic capacities, such as amino acid[56,57]. An example is the recently acquired facultative symbiont *Francisella-LE* (1.56 Mb) in *Amblyomma* ticks, which carries a larger genome with potential for higher biosynthetic capability than the obligate symbiont *Coxiella-LE* (0.66 Mb), showing a tendency of replacement with *Coxiella-LE* since *Francisella-LE* may supply additional benefits compared to *Coxiella-LE*[56]. Similarly, in *D. gallinae*, DGB2 encodes a greater number of nutrient synthesis pathways compared to obligate symbionts, such as *Bartonella* sp. DGB1[39] and *Rickettsiella*[44], which may improve or supplement host nutrient acquisition, particularly under nutritionally limited or stressful conditions. Based on genomic analysis (i.e., no substantial genome reduction and several complete nutrient biosynthetic pathways encoded), we propose that the dominant symbiont, *Bartonella* sp. DGB2, may act as a nascent non-obligate symbiont that plays a substitutive role in nutrient provision.

Besides vitamins and cofactors, iron represents an essential nutrient for hematophagous arthropods. For instance, *A. aegypti* can acquire iron from both heme and transferrin for embryonic development, with heme comprising the predominant source of bioavailable iron[8]. This process is dependent on HO catalysis, which degrades heme to release iron ions for subsequent utilization. However, although *A. aegypti* encodes functional HOs, their primary role is to alleviate oxidative stress through heme degradation rather than mediate iron metabolism. As such, the exact mechanism of heme iron acquisition in this species has not been fully elucidated. As outlined in the introduction, the acquisition of iron in ticks is also ambiguous, as they lack an HO-encoding gene. Similarly, the iron source of *D. gallinae* remains unexplored. The genome analyses of *D. gallinae* also revealed the absence of an HO-encoding gene[18], leaving the pathways underlying iron acquisition and utilization from heme elusive. Given that bacteria can obtain iron via heme uptake and degradation, we analyzed the heme uptake and degradation systems encoded by the three dominant bacteria (including *Bartonella* sp. DGB1, *Bartonella* sp. DGB2, and *K. massiliensis*) in *D. gallinae*, and identified a complete heme metabolic operon (the *hmu* operon) only in the DGB2 genome, which has been confirmed to facilitate bacterial acquisition of heme-derived iron from the host[45,58]. Subsequent observations revealed that DGB2 exhibited enhanced heme-binding capacity under iron deficiency stress, while exogenous heme supplementation rescued its growth defects. This pattern is similar to the iron utilization strategy reported in *M. tuberculosis*[59], further confirming that DGB2 can utilize heme as an iron source, and transcriptional analysis indicates that the *hmu* system likely mediates this process.

The *hmu* system imports heme via outer membrane receptor HmuR, followed by cytoplasmic heme degradation catalyzed by HmuS to release ferrous iron for cellular metabolism. This pathway represents a key iron-acquisition mechanism in numerous bacterial species[60]. In the present study, *E. coli* BL21 overexpressing HmuR from DGB2 exhibited enhanced heme-binding capacity, while HmuS displayed in vitro hemin-degrading activity. These results demonstrate that, despite the low amino acid similarity between HmuR/HmuS from DGB2 and other well-characterized homologous proteins, they retain high functional conservation and serve as core components that mediate heme-iron acquisition in DGB2. Additionally, an MbfA-type ferrous ion efflux pump (WP_008037385.1) was identified in the DGB2 genome, with transmembrane structures and iron-binding motifs homologous to those in *A. tumefaciens*, whose iron efflux activity has been experimentally verified[61]. Unfortunately, despite our best efforts, we were unable to generate a knockout mutant of this gene, which prevented us from verifying its function in DGB2 iron excretion. Nevertheless, the present study demonstrates that DGB2 harbors a complete heme uptake and degradation system that releases ferrous iron from heme and exhibits ferrous iron efflux capacity, indicating its potential to supply iron to the mite. To further verify the role of DGB2's iron supply to mites, we subsequently conducted an in vivo assessment.

We established a dysbiosis model via antibiotic treatment to investigate whether heme iron acquired by DGB2 can serve as one of the iron sources for *D. gallinae*. Antibiotic treatment is a widely used approach to characterize the functions of arthropod symbionts in insects and hematophagous arthropods[31,62,63]. For instance, tetracycline treatment can significantly inhibit *Coxiella* in *Haemaphysalis longicornis* (tick)[31], and a combination of ampicillin, gentamicin and cefotaxime can selectively eliminate *Hamiltonella* in the whitefly *Bemisia tabaci*[62]. In the present study, we performed a similar antibiotic treatment strategy and tested five antibiotics (amoxicillin, gentamicin, tetracycline, tobramycin and oxytetracycline) individually or in combination to screen for those capable of selectively inhibiting DGB2. At day 5 post-AMX treatment, DGB2 in mites was significantly inhibited. The other two dominant bacteria, *Bartonella* sp. DGB1 and *Kocuria* also exhibited slight fluctuations due to the broad-spectrum activity of the antibiotic, but showed no statistically significant differences overall. Although AMX treatment cannot completely exclude potential effects on some low-abundance bacteria, since nutritional symbionts (such as *Francisella* in ticks and *Wigglesworthia* in tsetse flies) generally represent the dominant taxa with high abundance[29,30,50], and low-abundance symbiotic bacteria have a low possibility of playing key roles, thus, such low-abundance bacteria in mites were not included in the further analysis. Selective elimination of DGB2 through antibiotic treatment significantly inhibited the reproductive capacity of *D. gallinae*, similar to the disruption of iron homeostasis induced by immune interference with the mite's ferritin[10]. As an essential micronutrient, iron has been well demonstrated to be critical for reproduction and embryonic development[6–8]. The present results showed that mites and their eggs exhibited significantly reduced iron content after DGB2-elimination, coupled with marked downregulation of the secreted ferritin DgFER1 and the intracellular ferritin DgFER2. These findings indicate that elemental iron availability is a limiting factor for this HO-deficient lineage (*D. gallinae*), and the heme uptake and degradation capacity of its symbiont DGB2 likely confers a significant reproductive advantage to the host.

Intracellular ferritins contain a conserved iron-responsive element (IRE) in the 5'UTR, enabling them to sense environmental iron levels and to serve as indicators of bioavailable iron, primarily storing excess iron ions within cells. Thus, downregulation of intracellular ferritin DgFER2 (referred to as FER1 in ticks) in DGB2-depleted mites indicates a decrease in intracellular bioavailable iron levels. By contrast, IRE is absent in secreted ferritins, which function more in iron transport or distribution to prevent cellular iron overload[11,64]. Interestingly, the expression level of secreted ferritin DgFER1 (also referred to as FER2 in ticks) was also significantly decreased, suggesting an indirect regulatory mechanism. Beyond post-transcriptional regulation mediated by IRPs and IREs, the expression of ferritin is also transcriptionally regulated by transcription factors, such as Nrf2[65]. Accordingly, we speculate that although secreted ferritin DgFER1 cannot directly sense iron levels, disturbed iron homeostasis (e.g., iron deficiency) in mites may activate other host feedback mechanisms (such as Nrf2) to regulate secreted ferritin expression. Similar results have been

observed in the expression of secreted ferritin HIFER2 (also without IRE) of *H. longicornis* (tick)[66].

In *H. longicornis*, ferritins were found to contribute crucially to reproductive function by regulating ovarian iron metabolism[66]. Similarly, our findings support that reproductive dysfunction in mites is closely associated with disrupted iron homeostasis. Specifically, DGB2 may serve as a potential pathway to provide bioavailable iron for *D. gallinae*, thereby profoundly affecting the mite's iron homeostasis and participating in the regulation of mite reproductive processes through ferritin. In addition, the possibility that DGB2 contributes to host reproduction by providing additional nutrients, such as B vitamins, cannot be ruled out. Although validation by rescue experiments via microinjection or feeding is important, such experiments are very technically difficult to perform in *D. gallinae* due to the tiny body size and the low efficiency of in vitro blood feeding[67]. Notably, antibiotic treatment reduced DGB2-mediated iron supply to the host, decreasing the mite fecundity but not completely inhibiting oviposition, which suggests the possibility of other iron sources (for example, transferrin-derived iron) for sustaining basic physiological activities. As has been well-documented in prior studies, transferrin-bound iron is also an iron source that facilitates embryonic development in both *I. scapularis*[13] and *A. aegypti*[8]. Accordingly, the contribution of this iron source to iron metabolism in *D. gallinae* remains further investigated.

Biological control of pests and their transmitted pathogens targeting essential symbionts represents a promising approach[68] which is generally based on the crucial relationship between symbionts and their hosts. The most successful application involves disrupting reproduction and embryonic development via the maternally inherited endosymbiotic bacteria *Wolbachia*-induced cytoplasmic incompatibility in planthoppers[69] and mosquitoes[70]. Employing symbionts as vectors to express effectors was found to suppress *Plasmodium*[71] and arbovirus[72] infection and transmission in hematophagous vectors (e.g., *Anopheles sinensis*). Additionally, antibodies against symbionts (e.g., *Rhodococcus rhodnii*) could inhibit the growth of the symbiont, thereby impeding the triatomine bug's fitness, indicating that symbiont-based vaccines are viable pest control strategies[73]. As shown above, many pest control strategies have been developed by targeting essential symbionts, but few via vaccine and no one has focused on iron homeostasis-related symbionts. Given the significant role of DGB2 in iron homeostasis and reproduction of *D. gallinae*, an inactivated DGB2 vaccine was administered to chickens, which elicited robust antibody responses. During blood feeding, these host-derived antibodies are transported into the mite's intestinal tract, which coincides with the primary colonization site of DGB2[39], and are therefore highly likely to exert a direct effect on intestinal DGB2 and further disrupt its biological functions. In this study, a significant inhibitory effect on the fecundity of *D. gallinae* was observed after mites ingested blood containing DGB2 antibody. This method achieved comparable efficacy to the antibiotic (AMX) intervention. Compared with antibiotics, a huge superiority of a vaccine based on symbionts is that zero antibiotic residue, which is really critical for laying hens. As Dr. Apolline Maitre pointed out, hematophagous ectoparasites generally ingest large amounts of blood containing host antibodies, which can retain their immune functions within the midguts of arthropod vectors, interfering with the physiology of vectors[68]. It should be pointed out that the present vaccination trial was conducted mainly to demonstrate the feasibility of a hematophagous vector control strategy based on iron homeostasis-related symbionts, and further mechanisms need future investigation, such as whether whole-bacterium-based vaccines can elicit polyvalent antibodies against other bacteria and if iron homeostasis of mites was affected by the antibodies. Anyway, the results at least showed that a vaccine based on iron homeostasis-related symbionts can offer a promising control strategy for hematophagous vectors and their transmitted pathogens.

Conclusion

The present study demonstrates that *Candidatus* Bartonella xiongi, a symbiont of *D. gallinae*, plays critical roles in the iron homeostasis of *D. gallinae*, and contributes to the regulation of the mite's reproductive capacity. Biological control targeting this iron homeostasis related symbiont represents an effective intervention strategy. These findings reveal a novel potential pathway through which symbionts provide the essential metallic element, iron, to hematophagous arthropods and a promising strategy for the control of hematophagous arthropods by targeting the iron homeostasis-related symbionts.

Materials and methods

Mite rearing and collection

The mite population utilized in this research was initially obtained from a commercial poultry farm in Pinggu District of Beijing, China[74]. The mites were then reared on chickens in an artificial climate incubator (RXZ-500B-LED, Ningbo Jiangnan Instrument Factory, China) at 30 °C, 75% relative humidity (RH), with a light: dark photoperiod of 12:12 h for approximately 6 years. Trapping devices were positioned near the chickens at the bottom edge of the cage for mite collection[75].

Isolation and morphological characterization of *Bartonella*

To isolate *Bartonella* from *D. gallinae*, 15 adult females were placed in a 1.5 mL centrifuge tube and soaked in 75% ethanol for 3 h to remove surface bacteria. Afterwards, the mites were washed three times with sterile water and then homogenized within 300 μL of sterile 1× phosphate-buffered saline (PBS). After appropriate dilution (it is advisable that the number of colonies growing on the culture medium after 24 h should be less than 50), 100 μL of the diluted homogenate was spread onto Tryptone Soy Agar (TSA) supplemented with defibrinated sheep blood (5%, vol/vol) and incubated in a carbon dioxide incubator (BPN-50CH, Shanghai Yiheng Scientific Instrument Co., Ltd., China) at 37 °C under 5% CO_2 for 4–5 days. Translucent colonies were chosen based on the colony morphology of *Bartonella* for PCR amplification of the 16S *rRNA* gene, and the PCR products were validated by Sanger sequencing. Colonies belonging to *Bartonella* were then subcultured onto the same medium for purity. The purified strain was designated as DGB2 and stored in TSB broth containing 50% (v/v) glycerol at −80 °C for further analysis.

Cell morphology of DGB2 was observed using transmission electron microscopy (TEM). For negative staining[76], the bacterial suspension was dipped into a copper grid coated with Formvar-carbon film, stained with 1% uranyl acetate for 3 min, and then observed with FESEM (SU-8010, Japan).

Whole-genome sequencing and comparative genomic analysis

Genomic DNA was extracted from DGB2 using the TIANamp Bacteria DNA Kit (Tiangen Biochemical Technology Co., Ltd, Beijing, China) and detected by agarose gel electrophoresis. Subsequently, the SMRT Bell library was established using the SMRT Bell™ Template Kit, and the Illumina sequencing library with an average insert size of approximately 350 bp was constructed using the NEBNext® Ultra™ DNA Library Prep Kit for Illumina (NEB, USA). Whole-genome sequencing of DGB2 was performed at Beijing Novogene Bioinformatics Technology Co., Ltd using PacBio and Illumina PE150 platforms. Low-quality reads were filtered by SMRT Link v8.0, following which the filtered reads were assembled using the Canu software (https://github.com/marbl/canu/) to generate a contiguous sequence without gaps.

For species-level taxonomic identification of DGB2, a comparison was carried out with all the genome sequences of type strains of species of the genus *Bartonella* (Supplementary Table 2) validly published under the ICNP, along with *Bartonella* sp. DGB1 isolated from *D. gallinae* in our earlier study[39] (designated *Bartonella* A in the original study), for further scrutiny. A whole-genome-based phylogenetic tree was reconstructed using the TYGS platform (available at: https://tygs.dsmz.de/)[77]. In silico DNA-DNA hybridization (DDH) values were calculated between the genomes of *Bartonella* species using GGDC software[78]. Average nucleotide identities (ANIs) were calculated using the IPGA online tool (available at: https://nmdc.cn/ipga/)[79].

Coding gene annotation and metabolic reconstruction

Protein-coding genes in the assembled DGB2 genome were predicted using the GeneMarkS program[80]. A whole-genome BLAST search (E-value less

than 1e-5, minimal alignment length percentage larger than 40%) was performed against the NR, KEGG, Pfam, and SwissProt databases for functional annotation of the coding sequence (CDS). Based on the genome annotation results, we reconstructed the pathways for the synthesis of B vitamins, cofactors, and EAAs using KEGG metabolic pathways as a reference, and subsequently evaluated their completeness and potential functionality. Furthermore, we also conducted a genome-level evaluation of the heme metabolism capacity of DGB2, including its ability for de novo heme synthesis[23] and the heme uptake and degradation systems[24].

Physiological and molecular responses under iron deficiency stress

To evaluate whether DGB2 can utilize exogenous hemin as the iron source for growth under iron-depleted conditions, a hemin utilization assay was carried out as previously described with minor modifications[59,81]. Specifically, strain DGB2 stored at −80 °C freezer was streaked on TSA blood agar plates and incubated for 4 days at 37 °C under 5% CO_2. A pre-culture was prepared by inoculating one colony into 5 mL of S10 liquid medium (Schneider's medium supplemented with 10% FBS and 5% sucrose), which contains no defibrinated sheep blood, and incubating for 3 days at 90 rpm under identical culture conditions. For the hemin utilization assay, three types of media were prepared: i) S10 medium (as control); ii) S10 medium supplemented with 100 μM ethylenediamine-N,N′-bis (2-hydroxyphenylacetic acid) (EDDHA, an iron chelator, to establish iron-depleted conditions); iii) S10 medium supplemented with 100 μM EDDHA and 5 μM hemin (as an iron source). Pre-cultures were re-inoculated into 5 mL of the aforementioned media, respectively, at an initial OD600 of 0.1, followed by incubation with shaking for 48 h. The proliferation of DGB2 was evaluated by quantifying colony-forming units (CFU) through plating serial dilutions of the cultures on TSA blood agar plates. Meanwhile, residual cultures were collected for subsequent analysis of transcriptional levels of heme metabolism-related genes.

The heme-binding capacity of DGB2 under iron deficiency stress was determined using a method outlined by Liu M et al.[82]. Hemin stock solution (100×) was prepared by dissolving 6 mg of hemin in 1 mL of 1 M NaOH. DGB2 cells cultivated in S10 medium or in an iron-deficient state were collected via centrifugation and subsequently resuspended in PBS buffer to adjust the OD600 to 1.0. Subsequently, the hemin solution at the working concentration was added to the bacterial suspension, and the resulting mixture was incubated at 37 °C under 5% CO_2 for 2 h with a rotational speed of 60 rpm. The bacterial suspension was centrifuged at 12,000 rpm for 2 min, and the heme-binding capacity was compared by measuring the absorbance of residual hemin in the supernatant at 405 nm.

To investigate the response of the heme uptake system to iron-deficient environments, total RNA was extracted from the cultures collected in the aforementioned hemin utilization assay using *AG RNAex Pro* Reagent (Aikerui Biological Engineering Co., Ltd., Hunan, China), followed by cDNA synthesis using *Evo M-MLV* RT Reaction Mix (Aikerui Biological Engineering Co., Ltd., Hunan, China). Subsequently, quantitative PCR (qPCR) reactions were performed to quantify the mRNA levels of heme receptor (*hmuR*) and HO (*hmuS*), using the SYBR Green *Pro Taq* HS Premix (Aikerui Biological Engineering Co., Ltd., Hunan, China) according to the manufacturer's instructions, with the relevant primers listed in Supplementary Table 4. All reactions were carried out on the StepOne™ Real-Time PCR System (Applied Biosystems). Amplification conditions consisted of denaturation at 95 °C for 10 min, followed by 40 cycles of 15 s at 95 °C and 1 min at 60 °C. Three biological replicates were included, and the 50S ribosomal protein L4 (*rplD*) served as the reference gene for normalization[83]. Relative expression of target genes was determined by the $2^{-\Delta\Delta CT}$ method with unprocessed cells as a calibrating condition.

Construction of the recombinant expression plasmid

To elucidate the roles of *hmuR* and *hmuS* in heme uptake and degradation, we conducted recombinant expression for subsequent in vitro functional validation. In-Fusion primers (Supplementary Table 5) were designed using the online tool (https://tool.vazyme.com:18002/cetool/simple.html) based on the full-length coding sequences of *hmuR* (GenBank accession number: WP_412058372.1) and *hmuS* (GenBank accession number:WP_412058371.1). Subsequently, PCR amplification was conducted using DGB2 genomic DNA as the template. Amplified products were purified with a gel recovery kit (DC301, Vazyme Biotech, Nanjing, China) and homologously recombined with EcoRI-digested linearized pET28a (for *hmuR*) and pColdI (for *hmuS*), according to the Uniclone One Step Seamless Cloning Kit (SC612, Genesand Biotech, Beijing, China) instructions. Recombinant plasmids were transformed into *E. coli* BL21 (DE3) competent cells, and transformants were plated on LB solid medium with antibiotics (Kana, 50 μg/mL for pET28a-*hmuR*; Amp, 100 μg/mL for pColdI-*hmuS*). Positive colonies containing pET28a-*hmuR* or pColdI-*hmuS* plasmids were identified through testing individual bacterial colonies by colony PCR.

Hemin binding assay of *E. coli* heterologously expressing hmuR

To evaluate the hemin-binding capacity of HmuR, HmuR overexpression was induced in a heterologous expression system as described before[84]. Specifically, overnight cultures of *E. coli* BL21 (DE3) cells containing pET28a-*hmuR* plasmids were inoculated into fresh LB/Kana liquid medium at a ratio of 1:100 and cultured at 200 rpm and 37 °C until reaching mid-log phase. Subsequently, 1 mM Isopropyl β-D-thiogalactoside (IPTG) was added to the cultures, and incubation at 37 °C for 6 h induced HmuR expression. Induced and uninduced pET28a empty vector, along with uninduced pET28a-*hmuR*, were served as controls and cultured under identical conditions. Induced and uninduced cells were harvested separately, washed three times with sterile PBS, and subsequently diluted to a final OD600 of 1.0. Hemin binding assay was performed following the above-described method applied to strain DGB2.

Heme oxygenase activity assay of HmuS

E. coli BL21(DE3) cells harboring the recombinant pColdI-*hmuS* plasmid were cultured overnight at 37 °C in LB/Amp medium. Six milliliters of the cells were then inoculated into 600 mL of fresh LB/Amp medium and cultured at 37 °C, 200 rpm until reaching an optical density (OD600) of 0.6-0.8. Protein expression was induced by addition of IPTG to a final concentration of 1 mM, and cells were incubated for an additional 20 h at 16 °C with shaking (140 rpm). Subsequently, cells were collected and subjected to ultrasonic crushing with a power (JY92-IIDN, Ningbo Scientz Biotechnology Co., Ltd., Zhejiang, China) of 300 W for 30 min (on for 3 s and off for 3 s). After centrifugation at 10,000 rpm for 15 min, the supernatant was collected, and the recombinant HmuS was purified using the His-Tagged Protein Purification Kit (CW0894S, Cowin Biotech, Jiangsu, China). Purified protein was dialyzed three times (6 h each time) against 50 mM Tris-HCl buffer (pH 7.4) to remove residual imidazole.

Heme binding assay was performed as previously described[85,86] with slight modifications. Briefly, apo-HmuS (20 μM) was added to a 96-well plate (200 μL per well), with the absorbance spectrum recorded at 300–750 nm on multifunctional microplate reader (Biotek, USA) before adding hemin. Aliquots of hemin (with increasing concentrations from 0 μM to 12 μM) were added into the wells, and the mixture was then co-incubated at room temperature for 1 h, after which the absorbance spectrum underwent a second measurement. Hemin-containing Tris-HCl buffer served as the negative control.

The HmuS-hemin complex was prepared according to the method described in a prior study[46]. Hemin was added to the purified HmuS to achieve a final hemin-to-protein ratio of 2:1. After 1 h of room-temperature incubation, the HmuS-hemin mixture was then applied to pre-equilibrated Sephadex G-25 dextran gel to remove noncoordinated heme. The column was eluted with 50 mM Tris-HCl, and protein peak at 280 nm was monitored via NanoDrop spectrophotometer (NanoDrop, USA). Upon elution of the protein peak from the desalting column, corresponding HmuS-hemin complex fractions were collected. The ascorbic acid-dependent degradation of hemin was then monitored[46]. Ascorbic acid at a final concentration of

5 mM was added to the HmuS-hemin complex as an electron donor, and spectral changes at 300-750 nm were monitored after 30 min of reaction. The HmuS-hemin complex without the addition of ascorbic acid served as the negative control. A biliverdin standard at a concentration of 0.005 mg/mL was included as a reference for comparison of characteristic absorption peaks of hemin degrading product (biliverdin).

Phylogenetic analysis and secondary structure prediction of MbfA

After verifying the functions of HmuR and HmuS, to evaluate whether the DGB2 strain is capable of exporting iron ions released from heme degradation, we conducted bioinformatics analyses on the ferrous iron efflux pump-encoding gene (WP_008037385.1) screened from its genome. The multiple sequence alignment was performed using ClustalW by aligning the candidate ferrous iron efflux pump sequence of DGB2 with a set of ferrous iron efflux transporters, such as MbfA (from *Rhizobium* spp., *Rhodobacteraceae bacterium*, and *Bradyrhizobium zhengyangense*), PfeT (from *Bacillus subtilis*), FrvA (from *Listeria monocytogenes*), and FieF (from *Escherichia coli*). Detailed information pertaining to these sequences is included in Supplementary Data 2. Phylogenetic relationships were inferred by the maximum-likelihood method (bootstrap replicates ×1000) using MEGA7 software. Conserved domains were identified by searching the NCBI Conserved Domain Database (https://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi) and visualized using Chiplot (https://www.chiplot.online/). Secondary structure and transmembrane domain topology of MbfA from strain DGB2 were predicted using DeepTMHMM (https://hpc.nih.gov/apps/deeptmhmm.html). Putative iron-binding motifs were identified by aligning sequences against the characterized iron-binding sites (E-6-Y and ExxH) from *A. tumefaciens*[61].

Antibiotic treatment and quantification of symbionts

To investigate the effects of DGB2 on the fitness of *D. gallina*e, an antibiotic-induced dysbiosis model of *D. gallinae* was established following previously described methods[39]. Preliminary experiments were conducted to screen for antibiotics that selectively inhibit DGB2, with five antibiotics—amoxicillin, gentamicin, tetracycline, tobramycin, and oxytetracycline—tested individually or in combination; only 5-day amoxicillin (AMX) treatment achieved selective inhibition of DGB2 without significant changes in the abundance of other dominant bacteria. Therefore, amoxicillin was used to induce a DGB2 dysbiosis model of *D. gallinae*. Based on the 3 R principles of animal welfare and considering the limitation of laboratory conditions, eight-week-old chickens were randomly assigned to two groups (control and AMX), with three replicates per group and more than 800 mites in each replicate. Amoxicillin sodium (Macklin, Shanghai, China) solution (200 mg/mL) was prepared in 0.9% NaCl. Chickens in AMX group received intramuscular pectoral injections of AMX at 180 mg/kg based on pre-experimental data, while control birds were injected with an equal volume of 0.9% NaCl. Based on the pharmacokinetic parameters of AMX in chickens[87], starved mite challenge was performed immediately after antibiotic injection. Each chicken was infested with 800 mites placed in feeding devices[88]. After 6 h of blood feeding, feeding devices were removed, and all engorged mites were recovered and transferred to an incubator (30 °C, 75% RH) for 5-day starvation, for subsequent analysis of bacterial abundance post-antibiotic treatment and a second round of blood feeding.

The starved mites were collected for bacterial abundance analysis, with thirty individuals designated as one biological replicate and three replicates per group. Mites were washed to eliminate any interference from surface microorganisms[39]. Total DNA was extracted using the SteadyPure Universal Genomic DNA Extraction Kit (Aikerui Biological Engineering Co., Ltd., Hunan, China). Subsequently, the dominant symbionts within *D. gallinae*, including *Bartonella sp*. DGB1 (described in our previous study), *Bartonella sp*. DGB2, and *Kocuria*[39], were quantified through qPCR. The amplification system and reaction protocol were adopted as described above, with the *actin* gene of *D. gallinae* serving as the internal reference. Primers were listed in Supplementary Table 4.

Effects of DGB2 inhibition on D. gallinae

Adult females 5 days post-AMX treatment were subjected to a second blood meal on antibiotic-free birds. Engorged individuals were recovered to evaluate the effects of antibiotic-mediated DGB2 inhibition on mite reproduction, blood digestion, and survival. Mites treated with the same process but without antibiotic treatment were used as controls.

Parameters employed to assess the reproductive capacity of mites include the oviposition rate, fecundity, and egg hatching rate. Specifically, 50 adult females (in one biological replicate) were placed in a 96-well plate, with each well housing a single mite, and the plate was then sealed with parafilm. Subsequently, the plates were incubated at 30 °C and 75% RH to facilitate oviposition, with the mites monitored under a stereomicroscope (SteREO-Discovery V12, Carl Zeiss, Jena, Germany) over a 5-day observation period. Six replicates were included for both control group and AMX group. Reproduction parameters were calculated as below[89].

Oviposition rate = No. of mites laying eggs / No. of mites × 100%

Fecundity = No. of eggs / No. of all mites (including non-ovipositing mites)

Egg hatching rate = No. of larvae / No. of eggs × 100%

The blood-digestion status of the mites was evaluated through two approaches[39]: i) observing appearance changes within the digestive tract under a stereomicroscope; ii) weighing the mites (100 mites per replicate, with three replicates established for each group) with a microbalance (Sartorius MSA125P-1CE-DI, Gottingen, Germany). Prior to infestation, the mites were weighed to calculate their average body weight. During 4 days subsequent to blood feeding, the mites were daily observed and weighed, and only the surviving individuals were incorporated into the statistical analysis. The impact of AMX treatment on adult females' digestion was quantified by calculating the digestion rate.

Digestion rate = (Average weight after feeding - Average weight after digesting period) /Average weight gain after feeding × 100%

Additionally, mortality rates of mites in the control and AMX treatment groups were also calculated, with three biological replicates (each consisting of 100 mites, housed in 9-cm disposable sterile Petri dishes and sealed with parafilm). The survival of mites was monitored daily for 5 days. Mites were deemed dead in the absence of active movement of legs evoked by either light or mechanical stimulation. Ultimately, the cumulative mortality rate of mites over a 5-day period was calculated for subsequent statistical analysis.

ICP-MS for iron in adult females and eggs

To explore whether DGB2 influences the reproductive capacity of *D. gallinae* through iron provision, inductively coupled plasma mass spectrometry (ICP-MS) was used to quantify iron content in AMX-treated mites and their eggs. To choose the optimized detection time when the interference from host hemoglobin-derived iron in undigested blood meal was minimized, the iron content in adult females was quantified at different time points (3, 4, 5, 6, 7, and 10 days) after blood meal. The engorged adult females were incubated at 30 °C and 75% RH. Subsequently, mites were collected at the specified time (with three biological replicates) and cryopreserved at −80 °C for further analysis.

Iron ICP-MS was conducted with reference to established protocols[8] with minor modifications. Briefly, samples were transferred into Mars X-Press Teflon digestion tubes (CEM Corporation, Matthews, NC, USA) and weighed for their mass. Subsequently, 5 mL of nitric acid was added to each vessel, and then placed into the MARS 6 microwave digestion unit (CEM Corporation, Matthews, NC, USA) for digestion. The group supplemented exclusively with nitric acid was designated as the negative control, with the objective of eliminating iron contamination originating from nitric acid and the diluent in subsequent analyses. Quantitative analysis of iron was carried out using the ICP-MS instrument (SUPEC 7000, Focus Technology, Hangzhou, China). The ICP-MS readings were expressed as ng Fe per mg of mites or eggs.

Adult females were treated with AMX as previously described, with 0.9% NaCl-treated mites as the control. After the second blood meal

(antibiotic-free blood), eggs laid by mites were collected; mites were sampled on day 7 (a time point with relatively stable iron content) for subsequent ICP-MS analysis, with three biological replicates per group.

Western blot analysis

Adult female mites from both the AMX-treated and control groups were collected at 48 h post the second blood meal. Total proteins were extracted from mite samples, separated by 10% SDS-PAGE, and then transferred onto a PVDF membrane. Primary antibodies specific to DgFER1 and DgFER2 (prepared and stored in our laboratory[10]) were used for western blotting, with GAPDH serving as the internal reference for tissue protein. A goat anti-mouse horseradish peroxidase (HRP) conjugate (TransGen Biotech, Beijing, China) was used as the secondary antibody. The signal was detected using a Tanon 5200 chemiluminescence detection system (Shanghai Tianneng Technology Co., Ltd., Shanghai, China).

Efficacy of inactivated bacterial vaccines

Given the significance of DGB2 to *D. gallinae*, an inactivated vaccine of DGB2 was prepared and administered to chickens to evaluate its protective efficacy against mites. Briefly, DGB2 was retrieved from TSA blood agar plates, washed three times with sterile 0.9% NaCl, and adjusted to a concentration of 5×10^9 CFU/mL. Inactivation was achieved with 0.3% formaldehyde (final concentration) added to the bacterial suspension, followed by shaking incubation at 37 °C and 150 rpm for 24 h. Inactivated bacteria were washed three times with 0.9% NaCl solution and resuspended to the initial volume. Six chickens were randomly allocated into two groups (adjuvant control and DGB2 inactivated vaccine-immunized), with three chickens per group. An aliquot of 400 μL of the bacterial suspension was homogenized with aluminum hydroxide adjuvant at a volume ratio of 6:1 and intramuscularly injected into chickens for immunization; controls were injected with the same volume of adjuvant. Booster immunizations were administered on days 14, 21, and 28 post-primary immunization. Blood samples were collected prior to each immunization (D0, D14, D21, and D28) and infestation (D35), and the separated serum was used for antibody titer determination.

Levels of DGB2-specific IgY antibodies were ascertained through enzyme-linked immunosorbent assays (ELISA) in accordance with prior research[90]. Heat-killed DGB2 antigens were prepared by heating bacteria in PBS at 56 °C for 1 h. Cells were collected and resuspended with carbonate-bicarbonate buffer (pH 9.6) and adjusted to a final concentration of 1×10^6 CFU/mL. A volume of 100 μL of antigen was coated onto a 96-well microplate and stored at 4 °C for an overnight duration. Plates were washed three times with PBST, and then blocked with 5% skimmed milk for 2 h at 37 °C. Diluted serum was added to the wells and incubated for 1.5 h at 37 °C. HRP-labeled goat anti-chicken IgG (Biosynthesis Biotechnology Co., Ltd., Beijing, China) was employed to detect bound antibodies. Color development was performed using TMB substrate solution (Solarbio Life Science, Beijing, China) and terminated by adding 50 μL of 2 M sulfuric acid. The optical density at 450 nm was read with a microplate reader (Bio-Rad Laboratories, Hercules, CA, USA).

To evaluate the protective efficacy of the DGB2 inactivated vaccine, adult females preconditioned by a 5-day starvation were employed for infestation with 800 mites per chicken. After 6 h of blood feeding, the feeding devices were removed, and all engorged mites were recovered. Subsequently, the reproduction, blood digestion, and survival of the mites were systematically evaluated following the described method above.

Statistics and reproducibility

All experiments included three or more independent biological replicates. Detailed information on sample sizes and number of replicates for animal and mite trials is documented in the corresponding Methods sections. Data analysis was performed using GraphPad Prism Software v8.0.1 (GraphPad Inc., San Diego, CA, USA) with the results expressed as the mean ± standard deviation (SD) of biological replicates. Student's *t* test or Mann-Whitney test was utilized for assessing differences across two groups depending on the data distribution, whereas one-way analysis of variance (ANOVA) coupled with Tukey's HSD post hoc test was applied to examine the variability among multiple groups. Statistical significance was determined using a cutoff P value of <0.05.

Ethics approval and consent to participate

All procedures and chick maintenance were conducted in accordance with the recommendations of the Guide for the Care and Use of Laboratory Animals, Ministry of Science and Technology, China, which was ratified by the Institutional Animal Care and Use Committee of China Agricultural University (Approval No.: AW40215202-2-10).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The datasets generated during and analyzed during the current study, including all source data, are available in the Supplementary Data files. Uncropped Western blot images are provided as Supplementary Fig. 9 in the Supplementary Information. Genome sequence of the strain DGB2 in this work has been deposited in the National Center for Biotechnology Information (NCBI) under the Bioproject accession number PRJNA1200254.

Received: 13 March 2026; Accepted: 3 July 2026;
Published online: 20 July 2026

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Acknowledgements

We sincerely thank Dr. Congli Yuan of Shanghai Jiao Tong University for his valuable assistance and constructive suggestions on this study. Graphical abstract was created with Biorender.com and stored at the following URL: https://BioRender.com/wtt3ufa.

Author contributions

T.-C.S. (Methodology, Data curation, Formal analysis, Visualization, Writing-original draft, Writing-review & editing); Q.L. and P.-L.W. (Methodology, Investigation, Formal Analysis); Y.Z. (Investigation, Writing-review & editing); W.-F.C. (Supervision, Methodology); D.-H.S., X.-D.G., and B.-H.W. (Investigation); B.-L.P. (Methodology, Supervision, Writing - Review & Editing, Funding Acquisition).

Funding

This work was primarily supported by the "National Natural Science Foundation of China" (Grant numbers: 32573387 and 32273027), Beijing Zhongnongda Veterinary Hospital Co., Ltd. and "Fundamental Research Program of Shanxi Province" (Grant number: 202503021212327).

Competing interests

The authors declare the following competing interests: A patent application related to the research content described in this manuscript was filed by China Agricultural University. B.-L.P., T.-C.S., Q.L., P.-L.W., Y.Z., and B.-H.W. are co-inventors. This study was conducted independently, with no influence from the patent applicant on data interpretation or conclusion formulation. All authors declare no other competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s42003-026-10671-5.

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Peer review information *Communications Biology* thanks Jan Hubert and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editors: Hannes Schuler and Tobias Goris.

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